

FYS4411/9411 April 10



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- Boltzmann machines (BM)
- Neural networks (NN)

ψ_T replaced by a NN or
BMs

From project 1 we have

$$\langle E(\alpha) \rangle = \int P(\vec{R}; \vec{\alpha}) E_L(\vec{R}; \alpha) d\vec{R}$$

$$\nabla_{\alpha} \langle E(\alpha) \rangle = 0 ; \text{ optimization}$$

BMS aka Energy models

$$P_i(\beta) = \frac{e^{-E_i \beta}}{Z(\beta)}$$

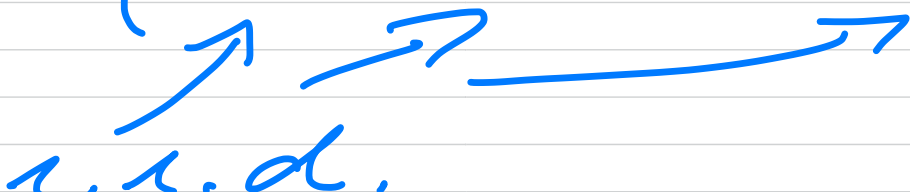
Boltzmann distribution

$$\beta = 1/k_B T$$

$$Z(\beta) = \sum_{E_i} e^{-\beta E_i} = 1$$

$$|4_T|^2 \sim \frac{e^{-E\beta}}{Z(\beta)} \leftarrow \text{parameter } \beta$$

$$X = \{x_0 x_1 \dots x_{n-1}\}$$



 i.i.d.

$$P(X) = \prod_{x_i' \in X} P(x_i')$$

assume that we have a set of parameters Θ which optimize $P(X)$ {MLE}

$$P(X) \rightarrow P(X; \theta)$$

we introduce latent variables / hidden variables

$$H = \{h_0, h_1, \dots, h_{m-1}\}$$

$$p(x; e) \rightarrow p(x, H; e)$$

Marginal probabilities

$$p(x; e) = \sum_{h_i \in H} p(x, h_i; e)$$

optimization problem

$$\hat{\theta} = \arg \max_{\theta \in \mathbb{R}^D} p(x; \theta)$$

$$p(x; \theta) = \prod_{x_i' \in x} p(x = x_i'; \theta)$$

$$\vec{\nabla}_{\theta} p(x; \theta) = 0$$

equivalently !

$$\hat{E} = \underset{\theta \in \mathbb{R}^P}{\operatorname{argmax}} \log p(x; \theta)$$

$$= \underset{\theta \in \mathbb{R}^P}{\operatorname{argmax}} \sum_i \log p(x_i; \theta)$$

or

$$\hat{E} = \underset{\theta \in \mathbb{R}^P}{\operatorname{argmin}} - \sum_i \log p(x_i; \theta)$$

$$p(x_i', h_j'; e) = \frac{f(x_i', h_j'; e)}{z(e)}$$

$$z(e) = \sum_{x_i', h_j'} f(x_i', h_j'; e)$$

$$P_i(\beta) = \frac{e^{-\beta E_i'}}{z(\beta)}$$

$$z(\beta) = \sum_{E_i'} e^{-\beta E_i'}$$

$$f(x, h; \theta) = e^{-E(x, h; \theta) / 4_T^2}$$

$$E(x, h; \theta) = ?$$

Binary - Binary $(R) \mathcal{B} M$

↑ restricted

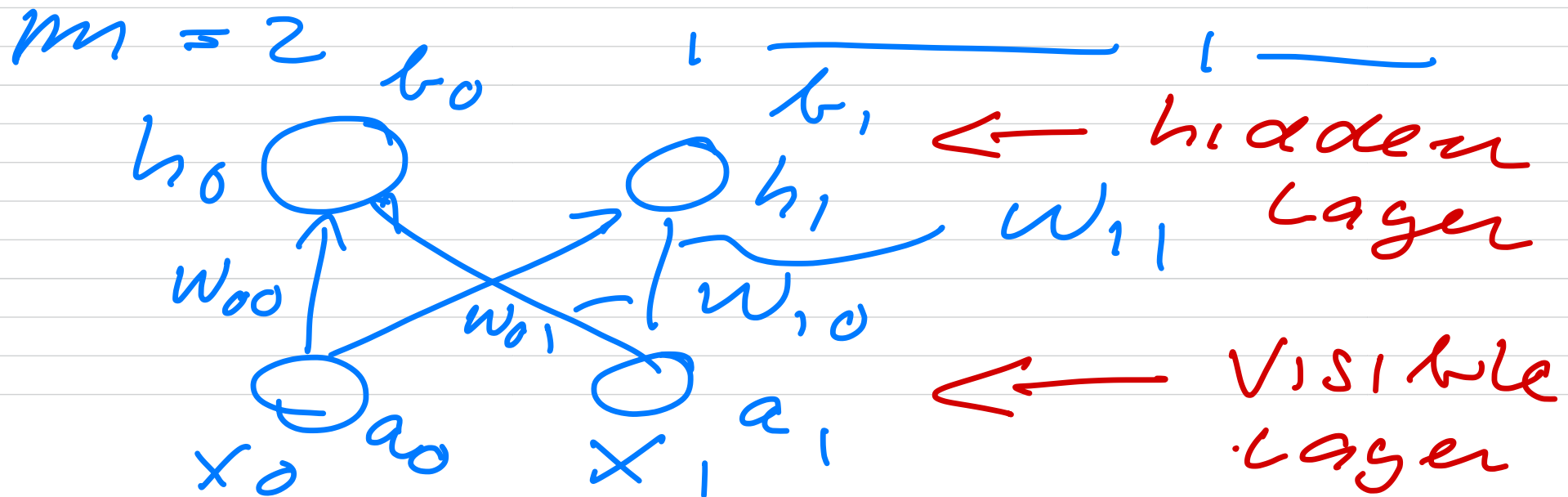
$$x_i = \{0, 1\} \wedge h_j = \{0, 1\}$$

$$E(x, h; \theta) = \sum_{i=0}^{n-1} x_i a_i + \sum_{j=0}^{m-1} h_j b_j + \sum_{i,j}^{n-1, m-1} x_i w_{ij} h_j$$

$$= \vec{a}^T \vec{x} + \vec{b}^T \vec{h} + \vec{x}^T W \vec{h}$$

$$\Theta = \{ \vec{a}, \vec{b}, W \}$$

$$n = 2 : \{0,0\}, \{0,1\}, \{1,0\}, \{1,1\}$$



optimization

$$\nabla_{\theta} \log p(x; \theta) = 0$$

$$= \nabla_{\theta} \left[\sum_{x_i \in X} \log f(x_i; \theta) \right]$$

$$- \nabla_{\theta} \log z(\theta)$$

$$\nabla_{\theta} \log z(\theta) = \frac{\nabla_{\theta} z(\theta)}{z(\theta)}$$

$$\frac{\nabla_{\theta} z(\theta)}{z(\theta)} = \frac{\sum_{x_i' \in X} \nabla_{\theta} f(x_i'; \theta)}{z(\theta)}$$

$$z(\theta) = \sum_{x_i'} \sum_{h_j'} f(x_i' h_j'; \theta)$$

$$f(x_i'; \theta) = \exp[\log f(x_i'; \theta)]$$

$$\frac{\nabla_{\theta} z(\theta)}{z(\theta)} = \frac{\sum_{x_i'} \nabla_{\theta} [\exp(\log f)]}{z(\theta)}$$

$$= \frac{\sum_{x_i'} \exp(\log f) \nabla_{\theta} \log f}{Z(\theta)}$$

$$= \frac{\sum_{x_i'} \overbrace{f(x_i'; \theta)}^{\text{red}} \nabla_{\theta} \log f(x_i'; \theta)}{\underbrace{Z(\theta)}_{\text{red}}, \quad \underbrace{p(x_i'; \theta)}_{\text{red}}}$$

$$= \sum_{x_i'} p(x_i'; \theta) \nabla_{\theta} \log f(x_i'; \theta)$$

$$= \mathbb{E}_{x_i \sim X} [\nabla_{\theta} \log f(x_i; \theta)]$$

cost/loss function

$$P(x_i, h_j; \theta) = \frac{f(x_i, h_j; \theta)}{z(\theta)}$$

Model

optimization

$$\nabla_{\theta} P(x, h; \theta) =$$

$$\nabla_{\theta} \log f(x, h; \theta) - \underbrace{E_{x \sim p_{h, \theta}}}_{\text{MC sampling}} [\nabla_{\theta} \log f(x, h; \theta)]$$

in project 2 (BM part)

cost function

$$\mathbb{E} [E_L(\theta)]$$

$$\nabla_{\theta} \mathbb{E} [E_L(\theta)]$$

$$|y_i|^2 = \frac{f(x; \theta)}{z(\theta)}$$

$$f(x; \theta) = \sum_h f(x, h; \theta)$$

$f(x, h; \theta)$

^{input}
:(i) Binary - Binary
RBM

(ii) Gaussian -
Binary
RBM

(iii) Gaussian -
Gaussian
RBM

⋮